

Assignment:

Design a Prototype of a Containerized Cloud-Based Application

Course: Cloud Architecture and Services

Max Marks: **15**

Course Outcome: CO6

Submission Deadline: **11th April 2026**

Assignment Type: Team of maximum 4 students from the same division

Note: Students should mention/use Free Tier services wherever applicable.

For evaluation viva/presentation/demo will be scheduled in the month of April 2026.

Objective:

To apply the concepts of cloud computing and containerization by designing a full-stack application architecture where the frontend and backend services are deployed using Docker containers on a cloud platform. The focus of this assignment is on system design and architecture, not full implementation.

Assignment Question

Select any real-world domain or application of your choice such as Healthcare, Agriculture, Education, Smart Cities, Banking, E-commerce, Tourism, Disaster Management, Transportation, or any other relevant area.

Design a cloud-based application architecture that solves a problem in your chosen domain. Your design must include frontend services, backend services, Docker containerization, and cloud infrastructure.

Your report should include the following sections.

1. Problem Identification

Describe a real-world problem in your chosen domain. Explain why a digital and scalable solution is required.

2. Application Overview

Explain the overall working of the proposed system.

Identify:

- Main users of the system
- Key functionalities provided by the application

3. Frontend Layer

Describe the frontend interface through which users interact with the system.

Mention possible technologies such as:

- HTML
- CSS
- JavaScript
- React or Angular

Explain the major features or pages available to users.

4. Backend Layer

Explain the backend services responsible for:

- Handling requests from the frontend
- Processing business logic
- Communicating with the database

Mention possible backend technologies such as:

- Node.js
- Python
- Java

Explain how the backend interacts with the frontend and database.

5. Containerization Using Docker

Explain how Docker containers will be used in the system.

Students should describe:

- Frontend container
- Backend container
- Docker images used to package the applications

Explain the advantages of containerization such as:

- Portability
- Consistent deployment
- Simplified application management

6. Cloud Infrastructure

Explain how cloud computing will support the system.

Describe how cloud services will be used for:

- Application hosting
- Database services
- Storage services
- Networking

Students may refer to platforms such as:

- Amazon Web Services
- Microsoft Azure
- Google Cloud Platform

Students should mention Free Tier services wherever possible.

Example:

AWS EC2 Free Tier instance, AWS RDS Free Tier database.

Students should also specify whether the database used is:

- Relational (SQL) such as MySQL or PostgreSQL
or
- Non-relational (NoSQL) such as MongoDB or DynamoDB

and justify their choice.

7. System Architecture Diagram

Draw a clear architecture diagram showing the interaction between system components.

Example flow:

User--> Frontend Application (Docker Container)--> Backend API (Docker Container)
--> Cloud Database--> Cloud Storage or additional cloud services

Students are encouraged to create the diagram using tools such as:

- Draw.io
- Lucidchart
- Excalidraw
- PowerPoint

8. Security Considerations

Explain how the system ensures security.

Students should briefly describe:

- Secure user authentication (for example login mechanisms or token-based authentication)
- Data protection (for example encrypted storage or controlled database access)
- Secure communication between system components (for example HTTPS or secure APIs)

Only conceptual explanation is required. Implementation is not required.

9. Scalability and Reliability

Explain how the system can handle an increase in users.

Students should briefly describe:

- Vertical Scaling
- Horizontal Scaling

Conceptual explanation is sufficient.

10. Advantages of Cloud and Container-Based Deployment

Explain the advantages of implementing the system using cloud computing and Docker containers.

11. Technology Stack Summary

Students must provide a Technology Stack Table summarizing the technologies used in the proposed system.

Students should mention/use Free Tier services wherever applicable.

Example format:

Component	Technology or Service Used	Purpose
Frontend	React / HTML / CSS / JavaScript	User interface
Backend	Node.js / Python / Java	API and business logic
Containerization	Docker	Packaging frontend and backend
Cloud Platform	AWS / Azure / Google Cloud	Hosting the application
Compute	AWS EC2 / Azure VM	Running containers
Database	MySQL / PostgreSQL / MongoDB	Data storage
Storage	AWS S3 / Azure Blob Storage	File storage
Authentication	JWT / OAuth	Secure login

12. Screenshots of the completed work with description

Attach screenshots to show the use of all technologies/services used with small description.

13. Individual contribution of each team member.

Submission Guidelines

Report must include complete steps from 1 to 13.

Font: Times New Roman

Font Size: 12

Line Spacing: 1.5

File Format: PDF

Students must submit original work. Copying reports or content from any source is strictly prohibited. In case of plagiarism, zero marks may be assigned to the entire team.